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# Engineering Circuit Analysis

Eighth Edition

## Chapter 11 AC Circuit Power Analysis

# Instantaneous Power

$$p(t) = v(t)i(t)$$

Resistor

$$p(t) = i^2(t)R = \frac{v^2(t)}{R}$$

Inductor

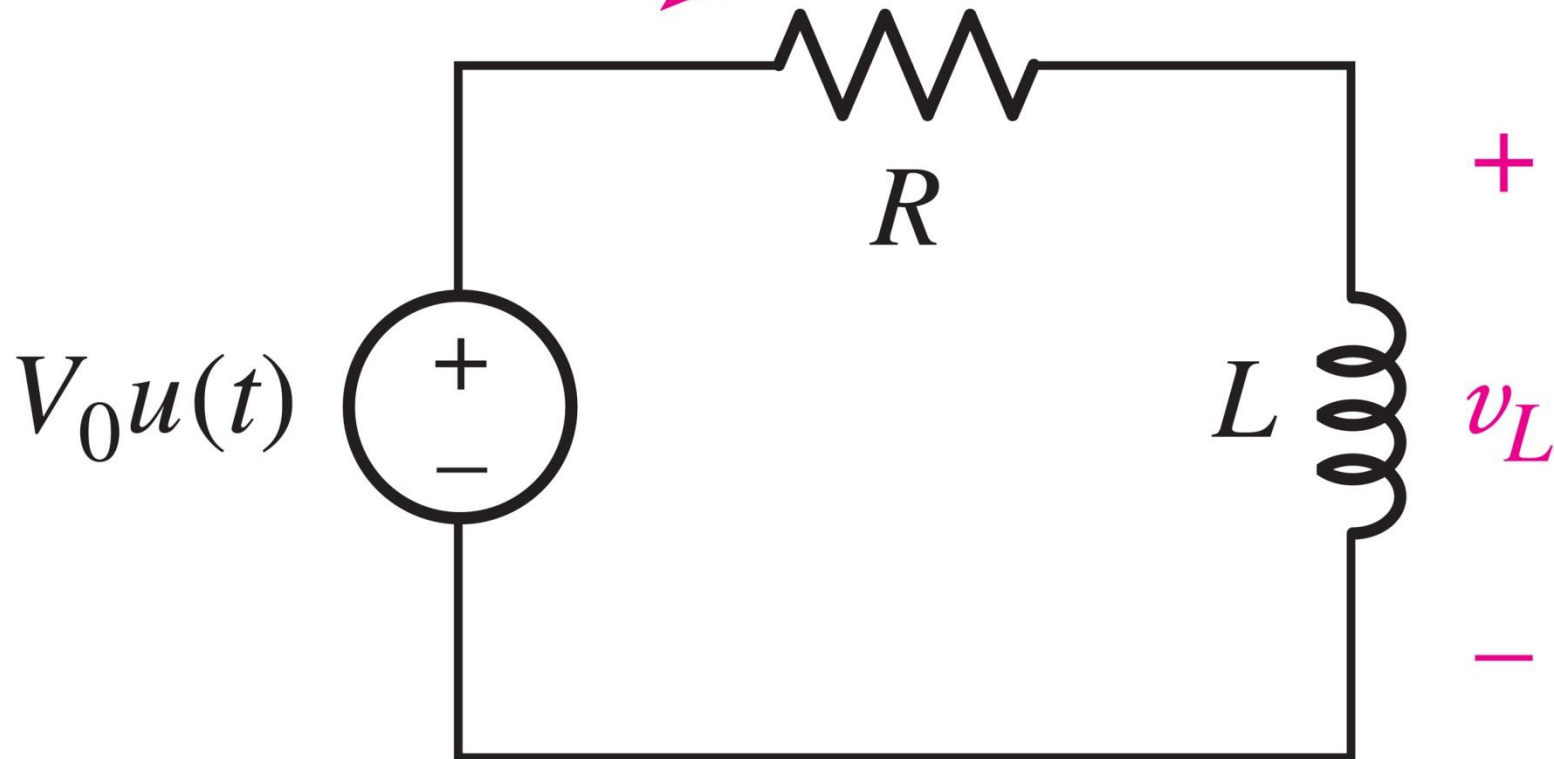
$$p(t) = Li(t) \frac{di(t)}{dt} = \frac{1}{L} v(t) \int_{-\infty}^t v(t') dt'$$

Capacitor

$$p(t) = Cv(t) \frac{dv(t)}{dt} = \frac{1}{C} i(t) \int_{-\infty}^t i(t') dt'$$

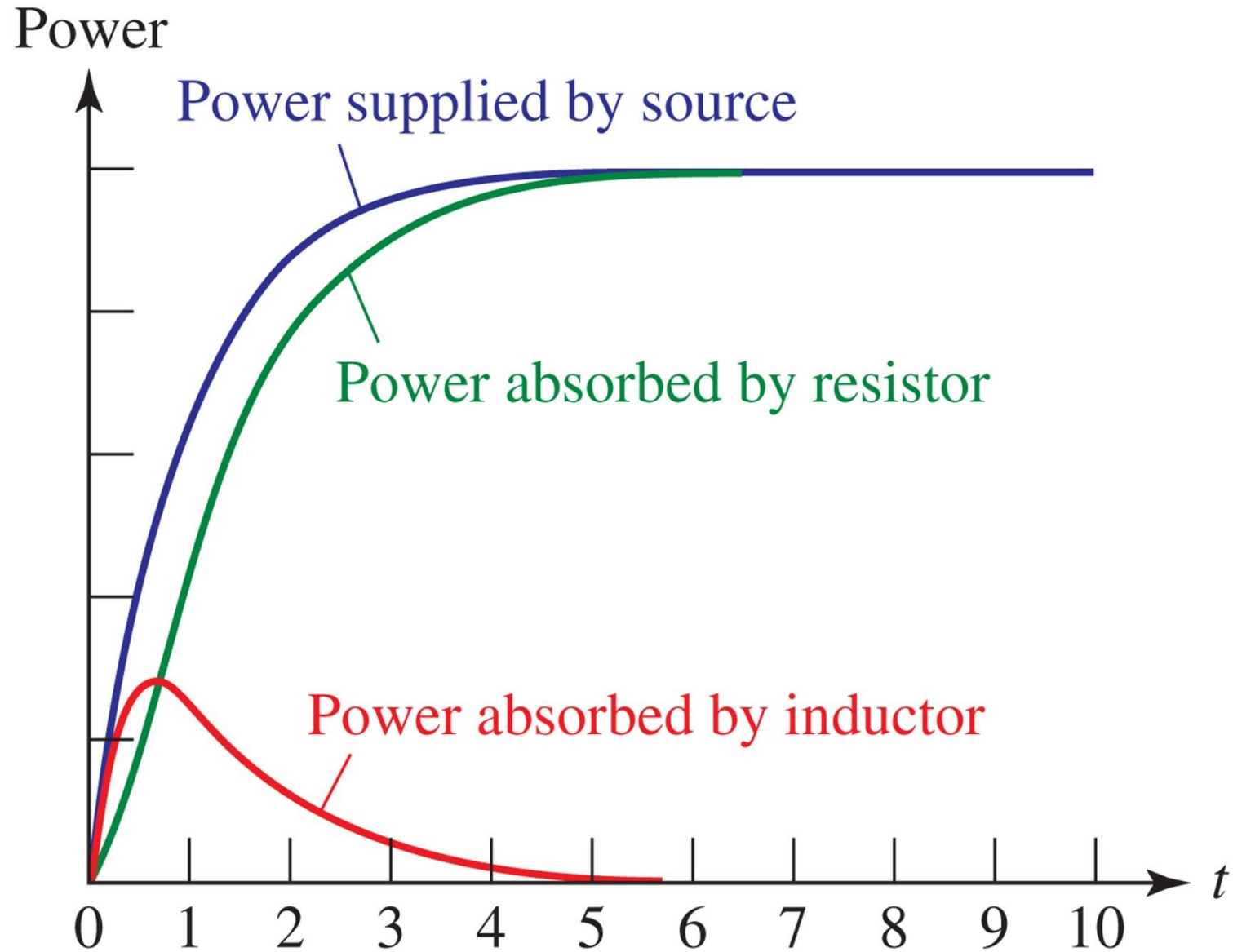
Consider simple R-L circuit

$$i(t) = \frac{V_0}{R} (1 - e^{-\frac{R}{L}t})$$



$$p_R(t) = i^2(t)R = \frac{V_0^2}{R} (1 - e^{-\frac{R}{L}t})^2 \quad p_L(t) = Li(t) \frac{di(t)}{dt} = \frac{V_0^2}{R} e^{-\frac{R}{L}t} (1 - e^{-\frac{R}{L}t})$$

$$p(t) = p_R(t) + p_L(t) = \frac{V_0^2}{R} (1 - e^{-\frac{R}{L}t})$$



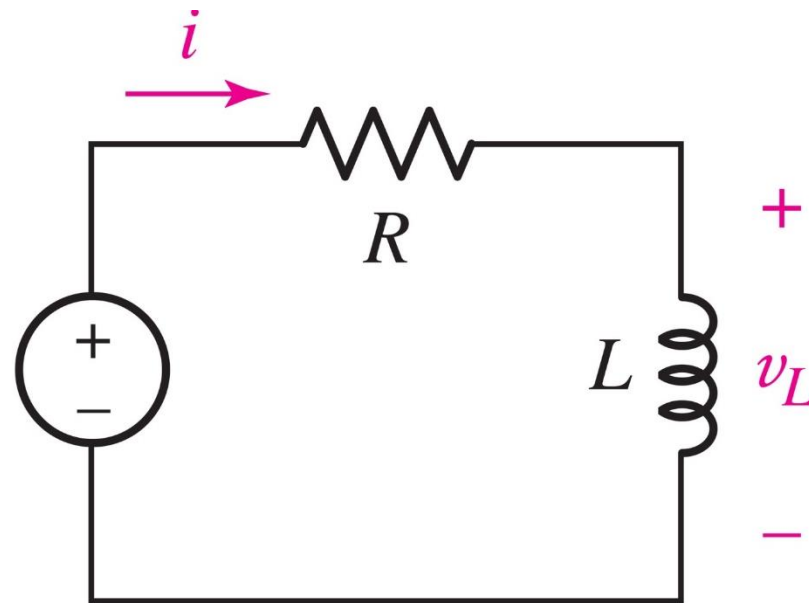
$$\sin \alpha \cos \beta = \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{2}$$

$$\cos \alpha \cos \beta = \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2}$$

$$\sin \alpha \sin \beta = \frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2}$$

Forcing  
function

$$v(t) = V_m \cos(\omega t + \theta)$$



Response

$$i(t) = I_m \cos(\omega t + \phi) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t - \tan^{-1} \frac{\omega L}{R})$$

$$p(t) = I_m V_m \cos(\omega t + \phi) \cos(\omega t + \theta) = \frac{V_m I_m}{2} \cos(\theta - \phi) + \frac{V_m I_m}{2} \cos(2\omega t + \theta + \phi)$$

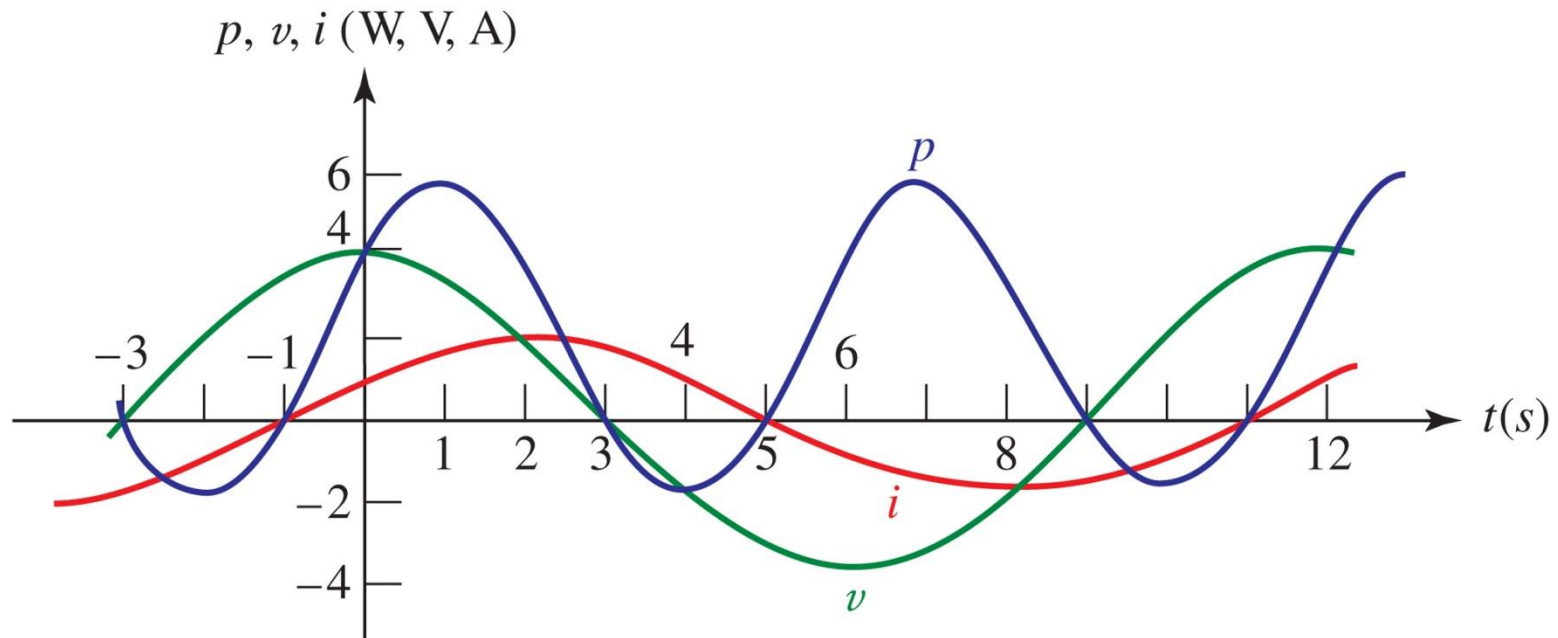
$$p_{av} = \frac{1}{T} \int_{t_1}^{t_1+T} p(t) dt = \frac{V_m I_m}{2} \cos(\theta - \phi)$$

**Ideal Resistor**  $P_{R,av} = \frac{1}{2} I_m V_m \cos(0) = \frac{1}{2} I_m V_m$

**Ideal Inductor or capacitor**

$$P_{C,av} = P_{L,av} = \frac{1}{2} I_m V_m \cos(90) = 0$$

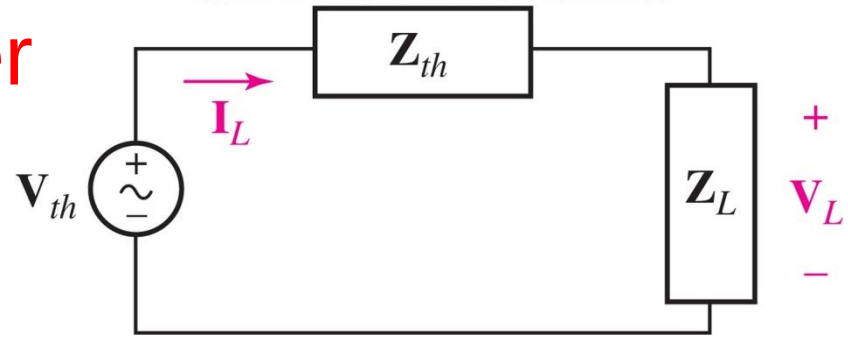
Find expression for instantaneous power and average power when  $\mathbf{V}=4\angle 0^\circ$  is applied across an impedance  $2\angle 60^\circ$ . The frequency  $\omega=\pi/6$



# Maximum Power transfer

$$Z_{th} = R_{th} + jX_{th}$$

$$Z_L = R_L + jX_L$$



## THEOREM

An independent voltage source in series with an impedance  $Z_{th}$  or an independent current source in parallel with an impedance  $Z_{th}$  delivers a maximum average power to that load impedance  $Z_L$ , which is the conjugate of  $Z_{th}$ , or  $Z_L = Z_{th}^*$

$$P_{av} = \frac{1/2 |V_{th}|^2 \sqrt{R_L^2 + X_L^2}}{(R_{th} + R_L)^2 + (X_{th} + X_L)^2} \cos \left( \tan^{-1} \frac{X_L}{R_L} \right)$$

# Effective Values

$$P_{R,av} = \frac{1}{T} \int_0^T i^2 R dt = \frac{R}{T} \int_0^T i^2 dt = I_{eff}^2 R$$

**Effective  
or root-mean-square  
(rms) values**

$$I_{eff}^2 = \frac{1}{T} \int_0^T i^2 dt$$

$$V_{eff}^2 = \frac{1}{T} \int_0^T v^2 dt$$

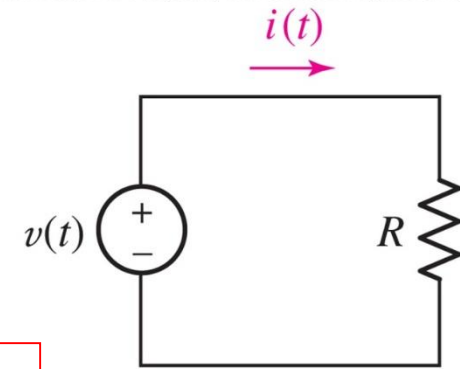
**For**

$$i(t) = I_m \cos(\omega t + \phi)$$

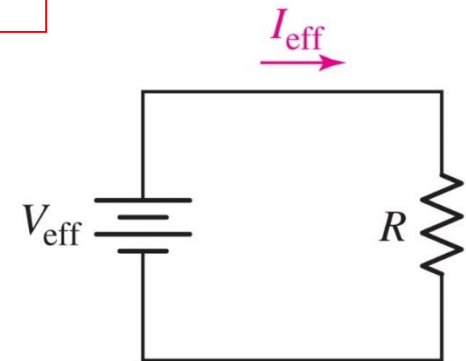
$$I_{eff} = \frac{I_m}{\sqrt{2}}$$

$$V_{eff} = \frac{V_m}{\sqrt{2}}$$

$$P_{av} = I_{eff} V_{eff} \cos(\theta - \phi)$$



(a)



(b)

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

**Apparent power**

$$P_{ap} = I_{eff} V_{eff} \quad [\text{VA}]$$

**Power factor  
(leading or lagging)**

$$PF = \frac{P_{av}}{I_{eff} V_{eff}} = \cos(\theta - \phi)$$

$(\theta - \phi) > 0$  Voltage leads the Current (L) - **lagging**

$(\theta - \phi) < 0$  Current leads the Voltage (C) - **leading**

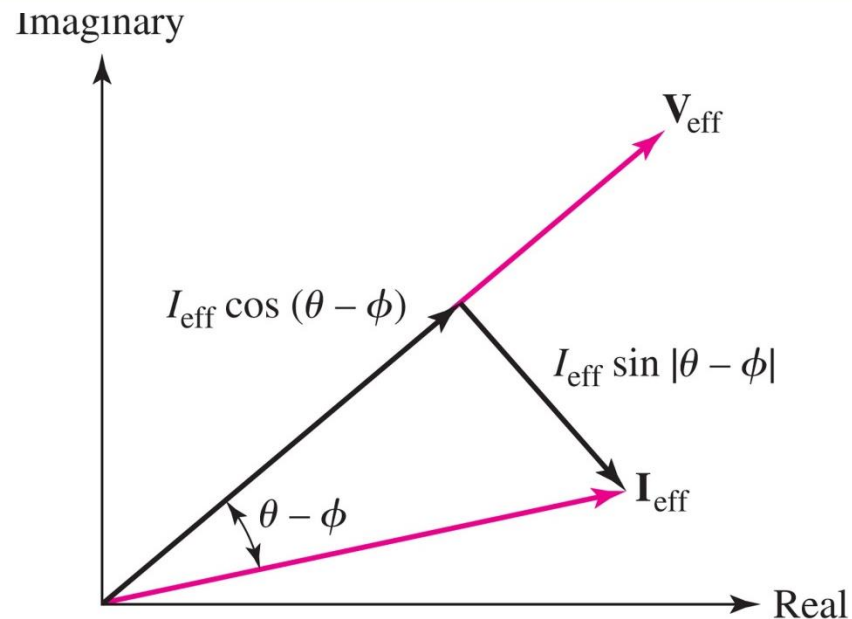
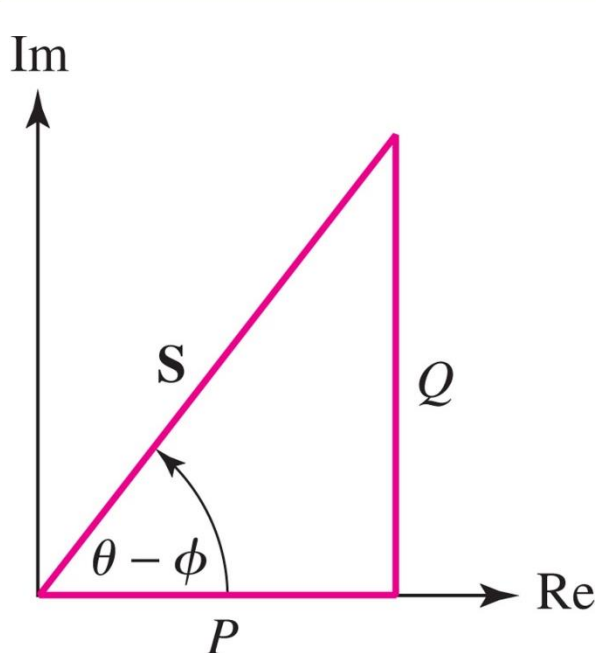
# Complex Power and Power Triangle

## Complex power

$$\mathbf{S} = \mathbf{V}_{\text{eff}} \mathbf{I}_{\text{eff}}^* = V_{\text{eff}} I_{\text{eff}} e^{j(\theta - \phi)} = P_{\text{av}} + jQ$$

TABLE 11.1 Summary of Quantities Related to Complex Power

| Quantity       | Symbol         | Formula                                                                                                                 | Units                      |
|----------------|----------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Average power  | $P$            | $V_{\text{eff}} I_{\text{eff}} \cos(\theta - \phi)$                                                                     | watt (W)                   |
| Reactive power | $Q$            | $V_{\text{eff}} I_{\text{eff}} \sin(\theta - \phi)$                                                                     | volt-ampere-reactive (VAR) |
| Complex power  | $\mathbf{S}$   | $P + jQ$<br>$V_{\text{eff}} I_{\text{eff}} \angle \theta - \phi$<br>$\mathbf{V}_{\text{eff}} \mathbf{I}_{\text{eff}}^*$ | volt-ampere (VA)           |
| Apparent power | $ \mathbf{S} $ | $V_{\text{eff}} I_{\text{eff}}$                                                                                         | volt-ampere (VA)           |



Calculate values of the average power delivered to each of two loads, the apparent power supplied by the source, and the power factor of the combined loads.

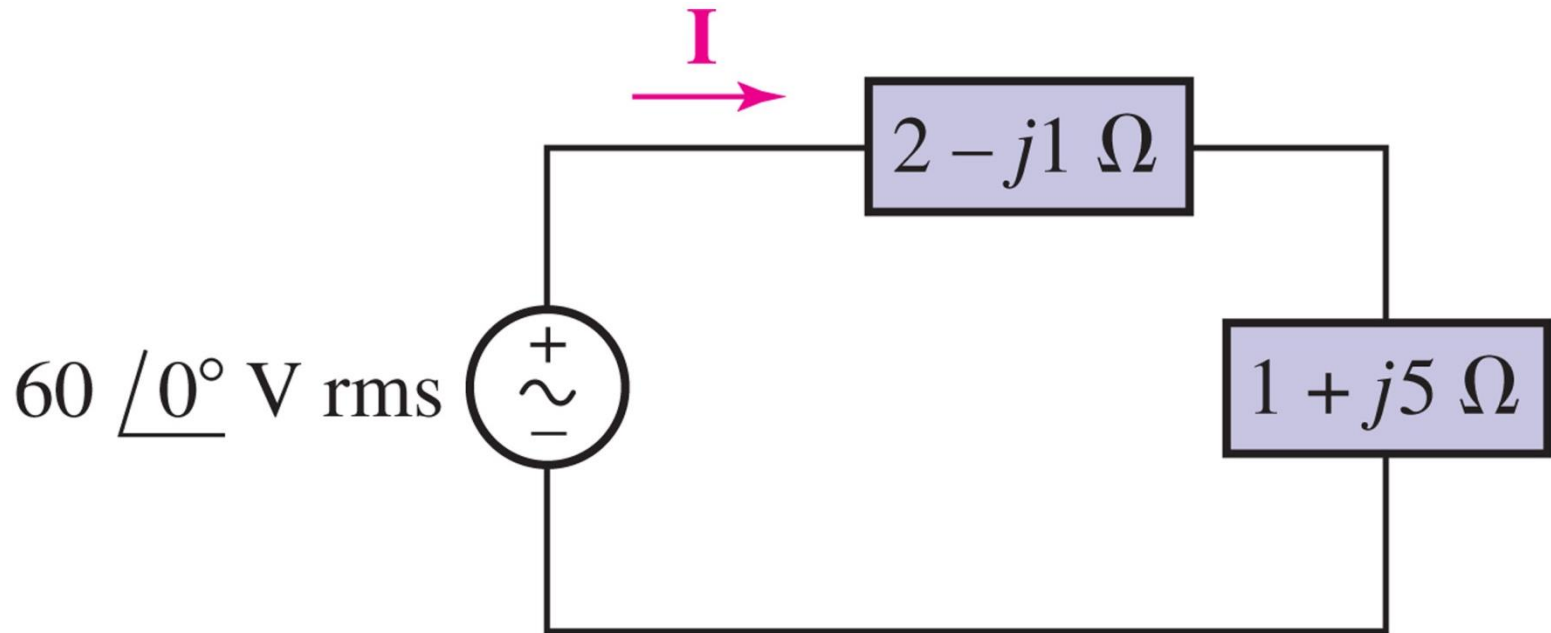


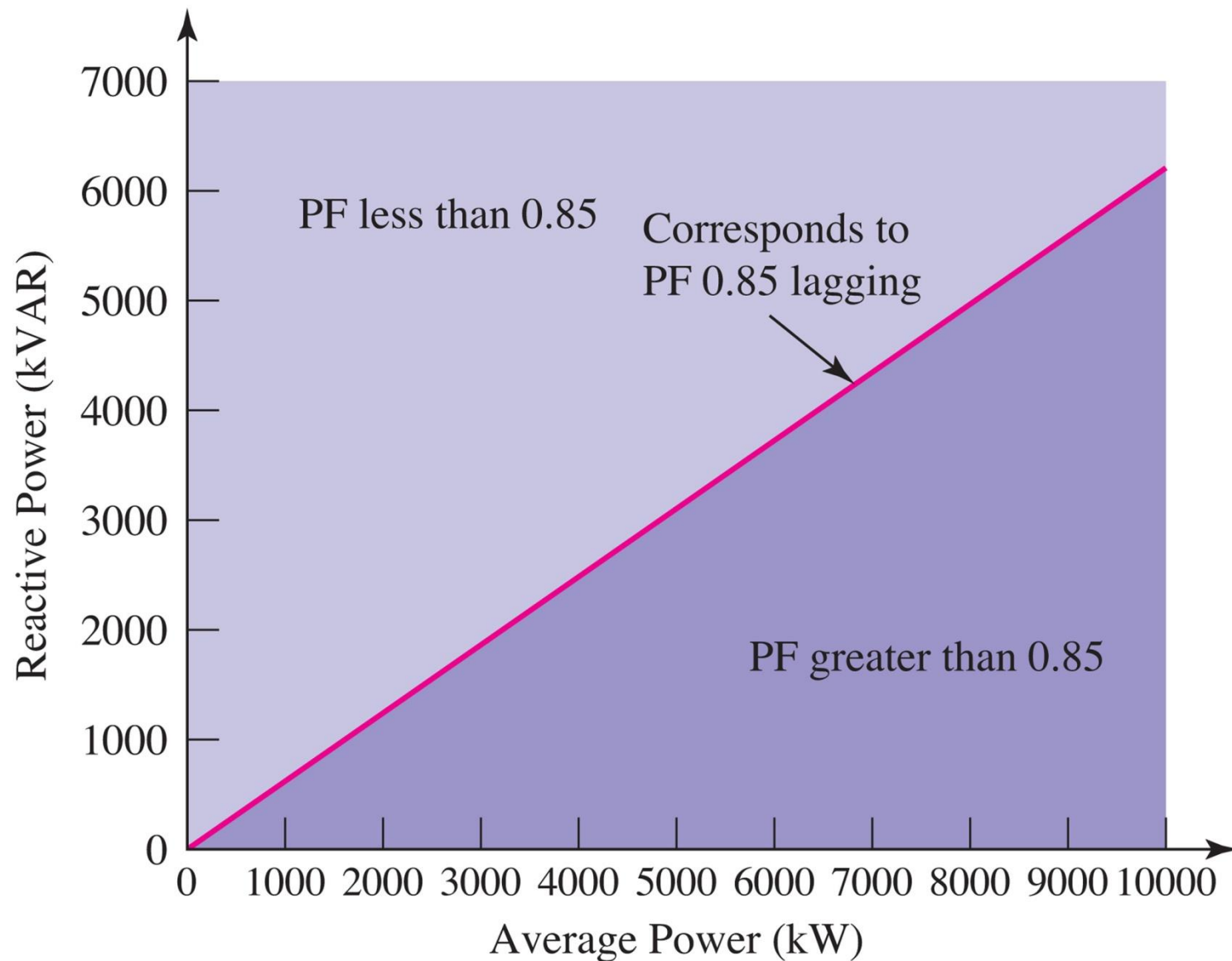
Fig. 1117

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Fig. 1118



$(\theta - \phi) > 0$  Current lags the voltage (L) - **lagging**

$(\theta - \phi) < 0$  Current leads the Voltage (C) - **leading**

**Fig. 1119**

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(Courtesy of Nokian Capacitors Ltd.)

An industrial consumer is operating 50 kW power motor at a lagging PF of 0.8. The source voltage is 230 V rms (60 Hz). In order to obtain lower industrial rate, the customer wishes to raise the PF to 0.95 lagging. Find corrective load.

